

Supplemental Material to the manuscript “Chiral Phonon Transport Induced by Topological Magnons” by

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RAYLEIGH-LIKE PHONON EDGE MODES

To describe the phonons, as discussed in the main paper, we consider a force constant model for out-of-plane phonon modes on the honeycomb lattice with only nearest neighbour interaction. This is described by the Hamiltonian

$$H = \sum_i \frac{p_i^2}{2m} + \frac{1}{2} \sum_{\langle i,j \rangle} C(u_i - u_j)^2, \quad (1)$$

where i and j are lattice site indices running over both the A and B sublattices of the honeycomb lattice.

To investigate edge modes in the system, we consider a finite ribbon geometry with periodic boundary conditions in one direction, and with a finite number of unit cells in the other direction. The edges of such ribbons can mainly be of two types: zigzag and armchair. The lattice geometries of these cases are shown in Fig. 1.

To find the phonon energy spectrum for these lattice geometries, we introduce the partial Fourier transform of the lattice site deviations and momenta, which for the lattice site deviation takes the form

$$u_{x,y}^D = \frac{1}{\sqrt{N_x}} \sum_k u_{ky} \exp(ik\hat{x} \cdot \mathbf{r}_n^D), \quad (2)$$

where $u_{x,y}^D$ is the lattice site deviation on sublattice D in unit cell (x, y) , $\mathbf{r}_{x,y}^D$ is the corresponding equilibrium position, and N_x is the number of unit cells in the horizontal direction. The periodicity requirement $u_{x,y}^D = u_{x+N_x,y}^D$ then gives $k = 2\pi n/N_x\lambda$, where λ is the periodicity of the lattice in the horizontal direction and n is an integer. For the zigzag edge ribbon, $\lambda = \sqrt{3}d$, while $\lambda = 3d$ for the armchair edge ribbon. This determines the size of the Brillouin zone.

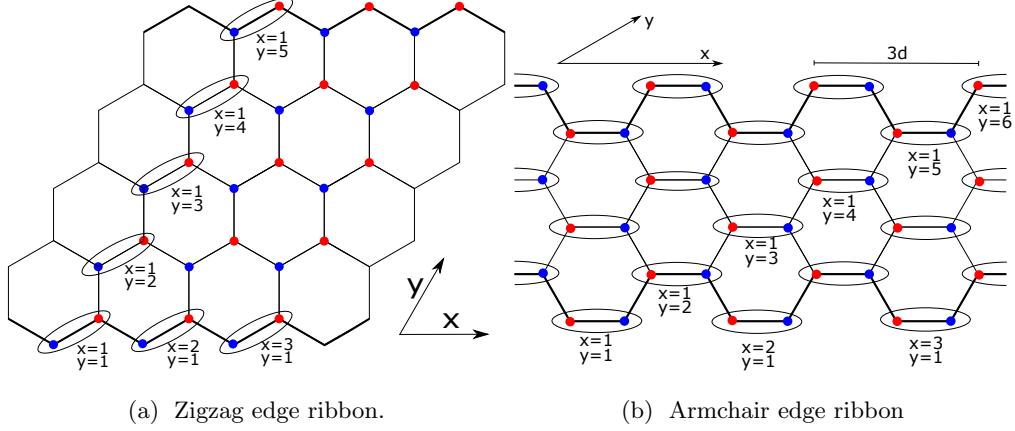


FIG. 1. Lattice geometries for the zigzag- and armchair edge ribbons, including unit cell labelling. We assume periodic boundary conditions in the horizontal direction and a finite number of hexagon layers in the vertical direction.

Introducing

$$u_k^\dagger = (u_{-k1}^A, u_{-k1}^B, u_{-k2}^A, u_{-k2}^B, \dots, u_{-kN_y}^A, u_{-kN_y}^B) \quad (3)$$

with similar notation for the momentum, the phonon Hamiltonian can be written on the form

$$H = \frac{1}{2m} \sum_k p_k^\dagger p_k + \frac{C}{2} \sum_k u_k^\dagger M_k u_k, \quad (4)$$

with a matrix M_k coupling the deviations on the various sublattices and neighbouring unit cell layers. This Hamiltonian is diagonalized through a unitary transform of the deviations and momenta followed by introducing phonon creation and annihilation operators $c_k^\dagger$ and c_k [1]. The excitation spectrum is then given by the phonon frequencies ω_{kn} , where $\omega_{kn}^2 = (C/m)\lambda_{kn}$ and $\{\lambda_{kn}\}$ are the eigenvalues of M_k .

Following this procedure for the ribbon geometry with zigzag edges, we obtain the spectrum in Fig. 2(a). The upper and lower branches of the phonon spectrum meet at $k_x d = 2\pi/3\sqrt{3}$ and $k_x d = 4\pi/3\sqrt{3}$, consistent with the result obtained by taking the 1-dimensional projection of the bulk bands. In Fig. 3, we plot the spatial profile of some selected eigenstates at quasimomentum $k_x d = 2\pi/3\sqrt{3}$. All the modes are delocalized.

Examining the armchair ribbon spectrum in Fig. 2(b), one may notice that the modes marked with green arrows stand out from the rest. If we were to compute the bulk spectrum and then perform

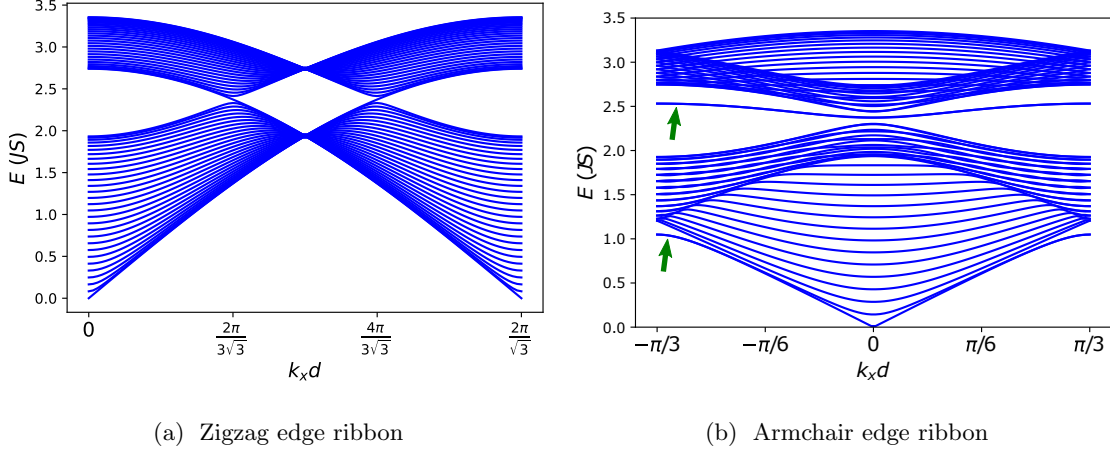


FIG. 2. One-dimensional projection of the energy spectrum for phonon modes on the honeycomb ribbon geometry with zigzag and armchair edges for $N_y = 30$ unit cells in the vertical direction (see Fig. 1). For the armchair ribbon, in addition to the bulk modes, there are edge modes marked with green arrows.

a 1D projection, the two modes marked in green would not be found. We therefore conclude that they must originate from an edge effect.

This is confirmed by examining the spatial profile of the modes, as shown in Fig. 4 for the modes between the upper and lower bulk phonon branches. The deviation amplitudes are finite on the outer armchair edges of the sample, and exponentially decaying into the interior of the sample. The inset shows the decay length as function of the inverse quasimomentum, and demonstrates that $\xi \propto 1/k_x$. This is perfectly analogous to the behaviour of so-called Rayleigh modes on the surface of a three-dimensional material [2]. Our modes can therefore be characterized as one-dimensional analogs of Rayleigh modes.

From the above discussion, it follows that that while the armchair edges support edge modes, the zigzag edge does not. This is rooted in the fact that on the edge unit cells of the armchair ribbon, both atoms have 2 nearest neighbours. For the zigzag ribbon, one atom has 2 nearest neighbours, but the other has 3. Vibrations are therefore easier to excite on the edges of the armchair ribbon.

COUPLED MAGNETOELASTIC MODES IN ZIGZAG RIBBON

To compute the excitation spectrum for the model with coupled magnon and phonon modes, we first calculate the phonon and magnon edge modes for the uncoupled model. The phonon modes

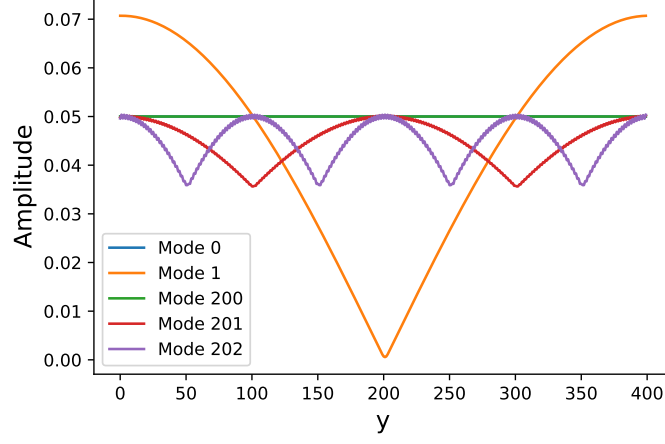


FIG. 3. Deviation amplitudes for selected zigzag ribbon phonon modes at $k_x d = 2\pi/3\sqrt{3}$ for $N_y = 200$ unit cells in the vertical direction. Fast oscillations have been averaged out. All modes are delocalized.

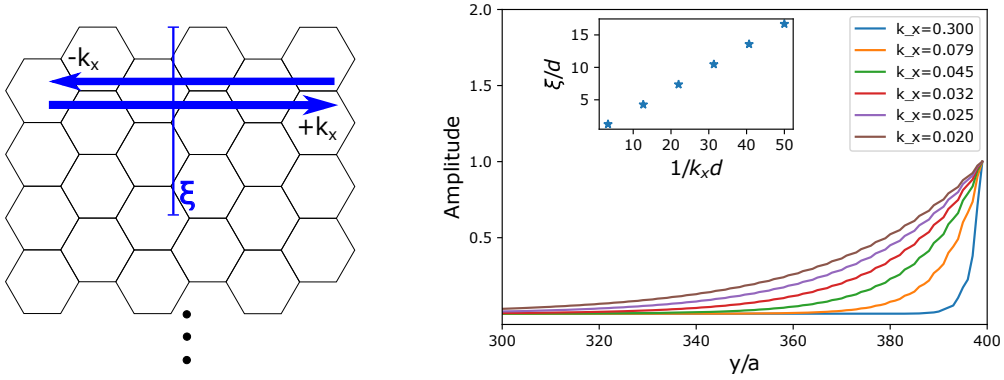


FIG. 4. (a) Rayleigh mode schematic. At a given armchair edge, two Rayleigh-like modes propagate along the edge with quasimomenta $\pm k_x$. The modes are localized within a distance $\xi(k_x)$ from the edge. (b) Deviation amplitudes for the Rayleigh-like edge modes as function of the vertical position y in units of the Bravais lattice constant a for different momenta k_x . The amplitudes are normalized to the value on the edge for easier comparison. The inset shows the localization length $\xi(k_x)$ as function of the inverse quasimomentum, demonstrating that $\xi \propto 1/k_x$, consistent with the behaviour expected from ordinary Rayleigh modes.

were discussed in the previous section, and we refer to the literature for the magnon spectrum [3]. Expressing the magneto-elastic coupling in terms of these eigenmodes and diagonalizing the resulting matrix, we obtain the excitation spectra.

For the zigzag edge ribbon, the spectrum is shown in Fig. 5. All phonon modes are delocalized. In the inset, we show the hybridization of the chiral edge magnon mode with some of these delocalized

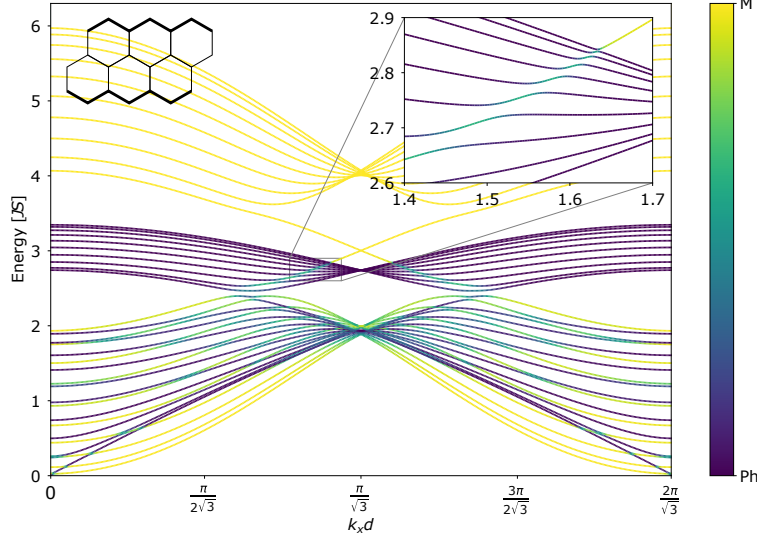


FIG. 5. Energy spectrum for the coupled magnon and phonon modes of the ribbon geometry with zigzag edge as function of quasimomentum k_x . The magnon (yellow) and phonon (purple) content of each mode is indicated with color.

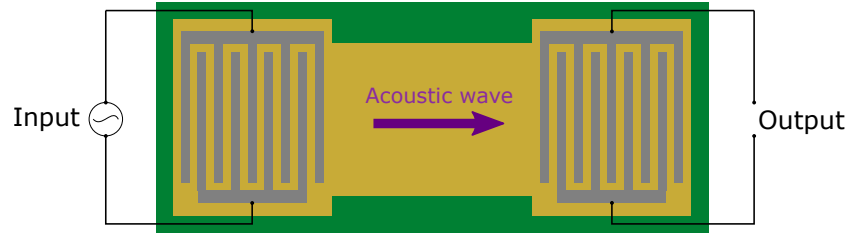


FIG. 6. Schematic depiction of an interdigital transducer employed to excite Rayleigh waves via an ac voltage on the left hand side. The same structure converts the acoustic waves back to ac voltage on the right and enables their detection. The metallic electrodes are lithographically patterned on top of a piezoelectric material, such as Lithium Niobate, into the depicted comb structure.

modes. The armchair ribbon spectrum has already been discussed in the main text.

INTERDIGITAL ELASTIC TRANSDUCERS

General principles and qualitative description

The interdigital transducer [4, 5] (IDT) consists of two metallic electrodes with a series of sections, called fingers, which are patterned into a comb-like structure on top of a piezoelectric material (Fig. 6). When a voltage is applied across the two electrodes, it creates a pattern of alternating charges on adjacent fingers via the capacitive effect. Via the constitutive properties of the piezo-

electric material, this results in a pattern of alternating strains. An applied ac voltage with angular frequency ω thus excites acoustic waves at the same frequency in the piezoelectric material. The wavelength is determined by the corresponding dispersion relation $\omega = ck$ [6], where c is the speed of sound in the material and k is the wavenumber. If the ensuing wavelength $\lambda = 2\pi/k$ is equal to the spacing between the adjacent fingers belonging to the same electrode, the acoustic signal interferes constructively and the excitation efficiency is high. If there is a mismatch between the finger spacing and the wavelength excited at the applied ac voltage frequency, the acoustic waves tend to cancel each other and excitation efficiency is low. With an increasing number N of fingers, the reinforcement or cancellation effect is stronger and the excitation resonances become sharper. Thus, the operation principle of an IDT is similar to that of a Bragg grating. Then, it is easy to understand that peaks in excitation are observed at multiple frequencies (and wavelengths) corresponding to the finger spacing being multiples of the acoustic wavelength. The fundamental peak is the strongest and subsequent overtones are progressively weaker as demonstrated by the frequency transfer characteristics discussed below.

Conventionally, IDTs have been employed in applications such as analog filters, and their desired operation frequency range has been from MHz to several tens of GHz [4]. A typical piezoelectric material employed is Lithium Niobate with a Rayleigh wave speed of 3.3 km/s. Thus, the fundamental peak corresponding to a center frequency of 1 GHz requires finger spacing of around 1 μm , which could easily be achieved via photolithography techniques. With contemporary electron-beam lithography techniques, a finger spacing of several tens of nanometers is readily possible, thus allowing a fundamental frequency of tens of GHz. Employing higher overtones allows pushing the operation frequency to several tens of GHz, and is predominantly limited by the driving electronics [5]. Due to the purview of their conventional applications, attempts to achieve higher frequencies have been limited. With recent advances in ultrafast lasers, several conventional methods have been adapted to achieve coherent phonon generation in the THz regime [7]. Thus, the operation range of the proposed method is estimated to be rather wide with up to hundreds of GHz in frequency and tens of nanometers in wavelength. The wavenumber selectivity can also be increased, in principle to arbitrary values, by using a large number of fingers. Combined with the tunability of the exact magnon-phonon anticrossing point (via an applied field, for example) across a broad range of frequencies and wavevectors, the proposed experimental method is well within the range of the contemporary state-of-the-art technology.

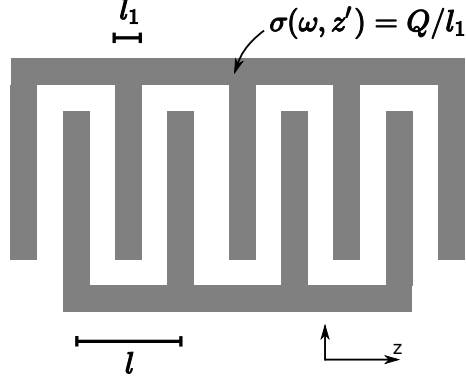


FIG. 7. Modeling the acoustic output of an interdigital transducer within the delta function model. An applied voltage generates charges on the metallic electrodes via the capacitive effect. The accumulated charge (or equivalently electric field) is converted into strain via the piezoelectric effect.

Frequency resolved acoustic output

To supplement the above qualitative discussion of the operating principles of an IDT, we now discuss its frequency resolved excitation efficiency within the so-called delta function model (Fig. 7). This model assumes that the charge accumulated on each finger is distributed uniformly and that the acoustic output is a linear superposition of the strain produced by the full charge distribution. In evaluating the strain (and thus the acoustic output) at a given point due to different fingers, the phase difference due to wave propagation from the fingers needs to be accounted adequately. The amplitude of the excited acoustic wave $A(z, \omega)$ at a position z is thus given by

$$A(z, \omega) = \alpha \int \sigma(\omega, z') e^{-ik(z-z')} dz', \quad (5)$$

where α is the charge-strain coupling factor of the piezoelectric material and $\sigma(\omega, z')$ is the charge density accumulated at position z' . The acoustic output of the IDT is thus simply the Fourier transform of the accumulated charge density. The square-wave like pattern of the accumulated charge on the comb-like structure thus suggests a sinc function response.

Referring the readers to detailed derivations and discussion in Ref. 4, we simply present the key result here. For an N -finger comb, the overall acoustic amplitude outside the IDT region becomes

$$A(\omega, z) = i\alpha Q \frac{\sin(kNl/2)}{\cos(kl/4)} \text{sinc}\left(\frac{l_1}{\lambda}\right) e^{ik(N-1)l/2} e^{i(\omega t - kz)}, \quad (6)$$

where Q is the charge on a single finger, l and l_1 are defined in Fig. 7, and $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. The equation above fully describes all the design characteristics of the device. Its further analysis

shows that the center or fundamental frequency ω_0 is determined by the condition $k_0 l = 2\pi$ (Note that $\omega_0 = ck_0$), while the bandwidth between the zeros in the response is given by $\Delta\omega/\omega_0 = 2/N$, in consistence with the qualitative discussion above.

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